Unnamed_Unit

Unit 2 - Introduction to Code Editor

PDF Documents

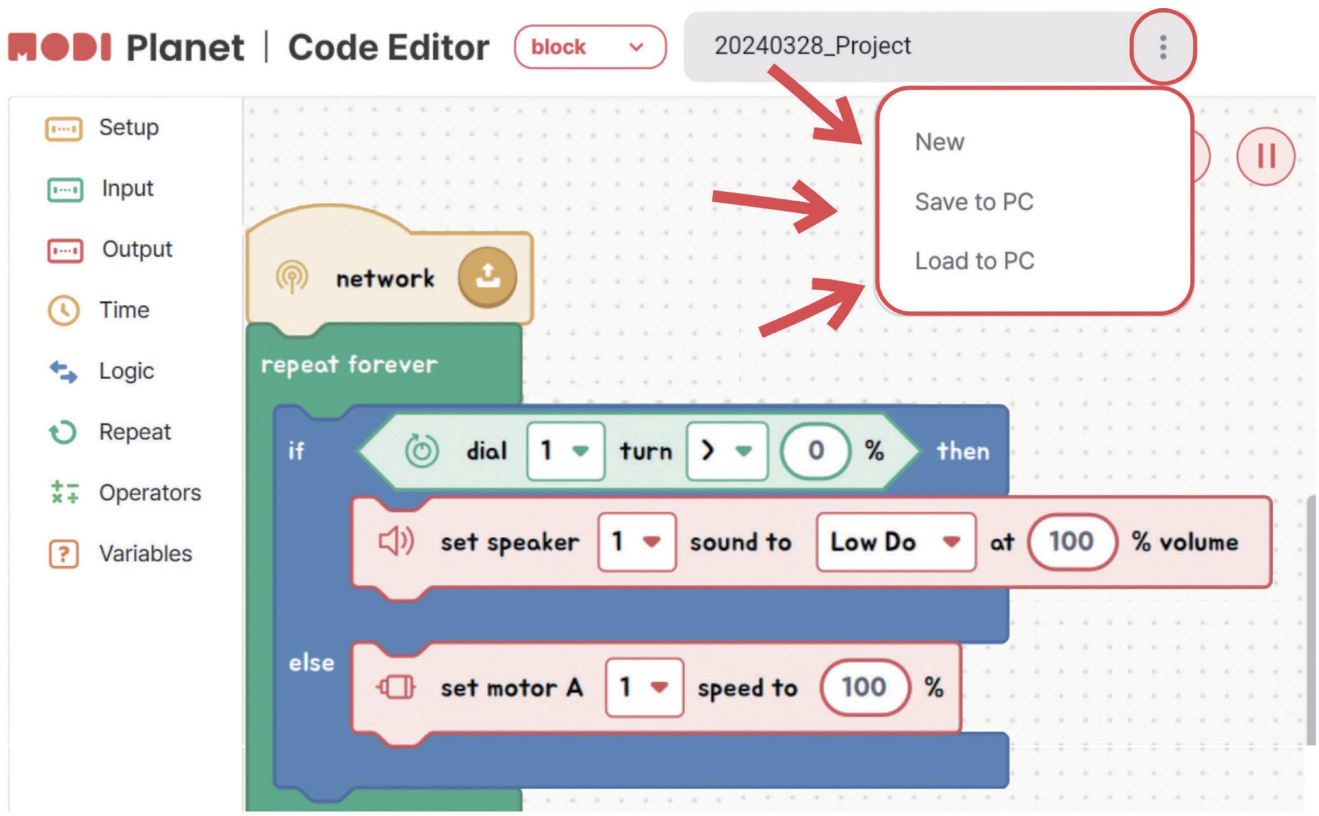
PDF Document 1: 17382188885426-Chapter 8 - Unit 2 - Introduction to Code Editor.pdf

This PDF document is included in the unit materials.

UNIT 2 INTRODUCTION TO CODE EDITOR

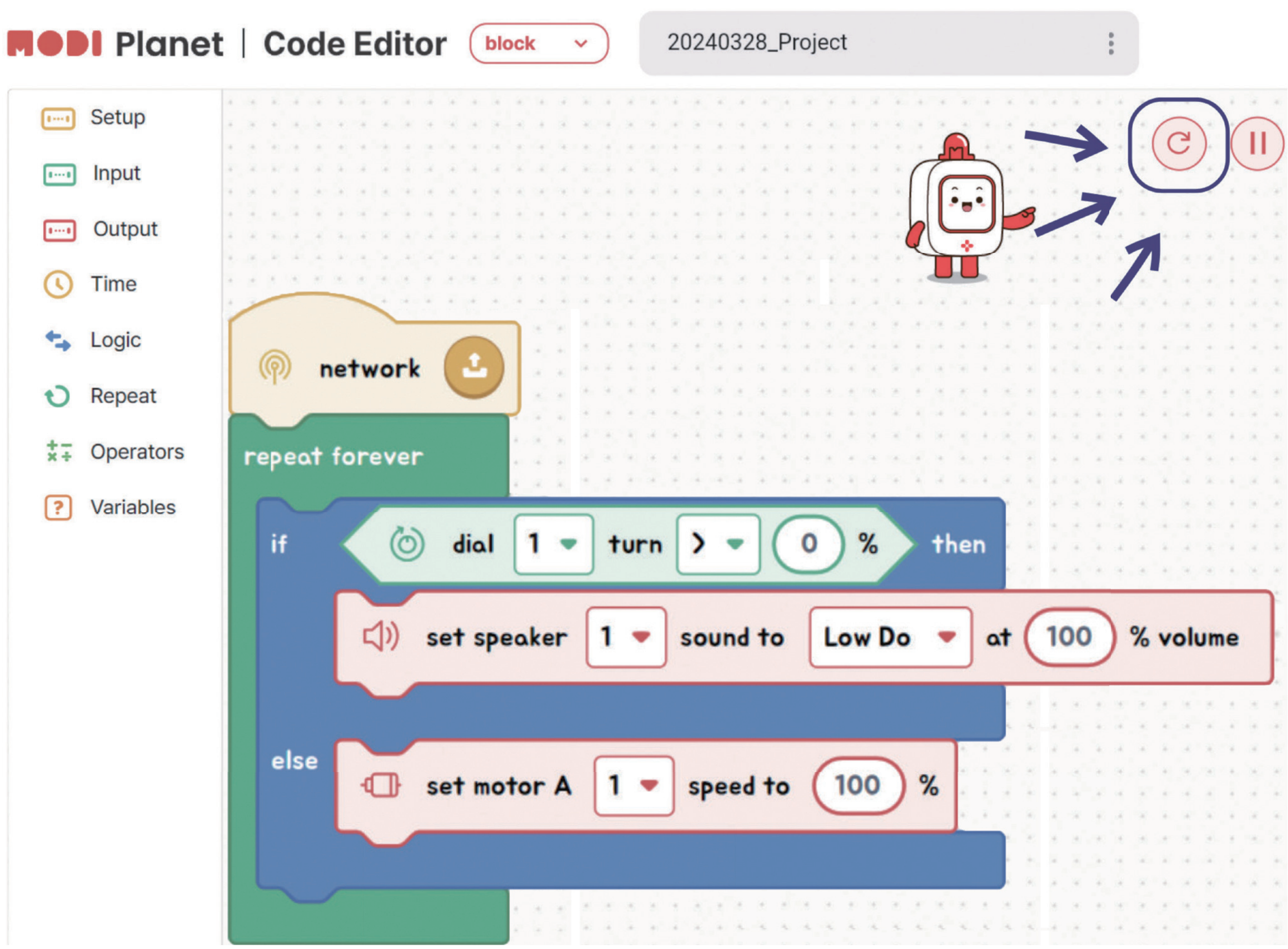
SAVE & LOAD PROJECTS

Click the ellipsis in the picture below to open a popup with options to save and load your projects.



INITIALIZING MODULES

1. Ensure the modules you wish to initialize are connected to your device.
2. Click the 'Initialization' button to reset them.

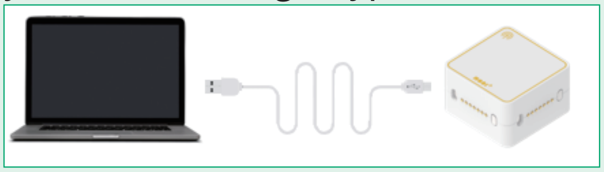


CONNECTING MODI TO DEVICE

Once the MODI Code Editor is set up, let's connect your MODI modules to your device.

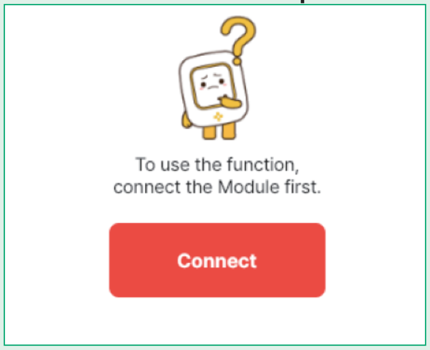
Step 1

Connect the Network Module to your device using a Type-C cable.



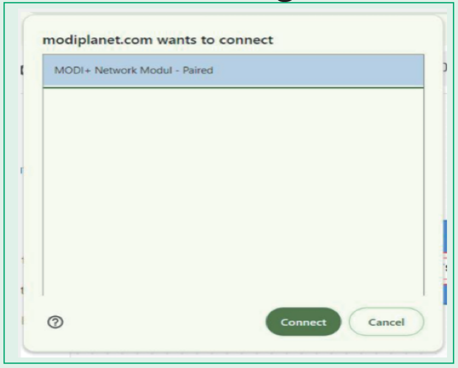
Step 2

Click the connect button in the module map.



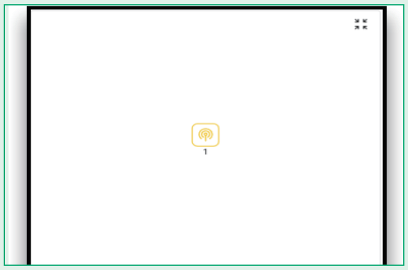
Step 3

Select 'MODI+ Network Module-Paired' and then click 'Connect' again.

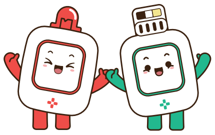


Step 4

A network icon will appear in the Module Map when the connection is successful.



Any additional modules connected alongside the Network Module will be displayed in the Module Map.



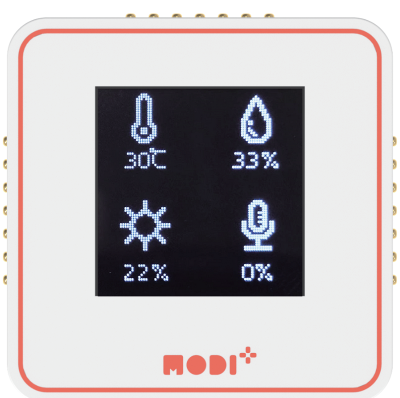
MONITORING MODULES

Monitoring is key to understanding a module's functions and current state. Always monitor new modules upon connection.

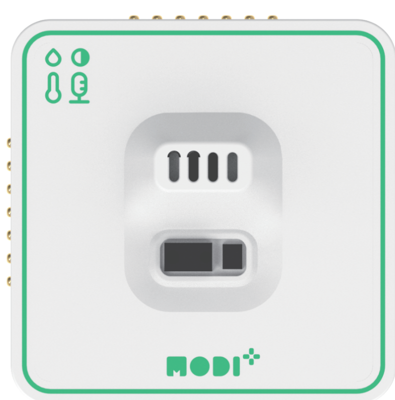
Mornitoring Input Data

Now, let's monitor the input data from the Environment Meter we built in Unit 1.

Do you remember the meter device shown below? Make sure the icons for its modules are visible in the Module Map!



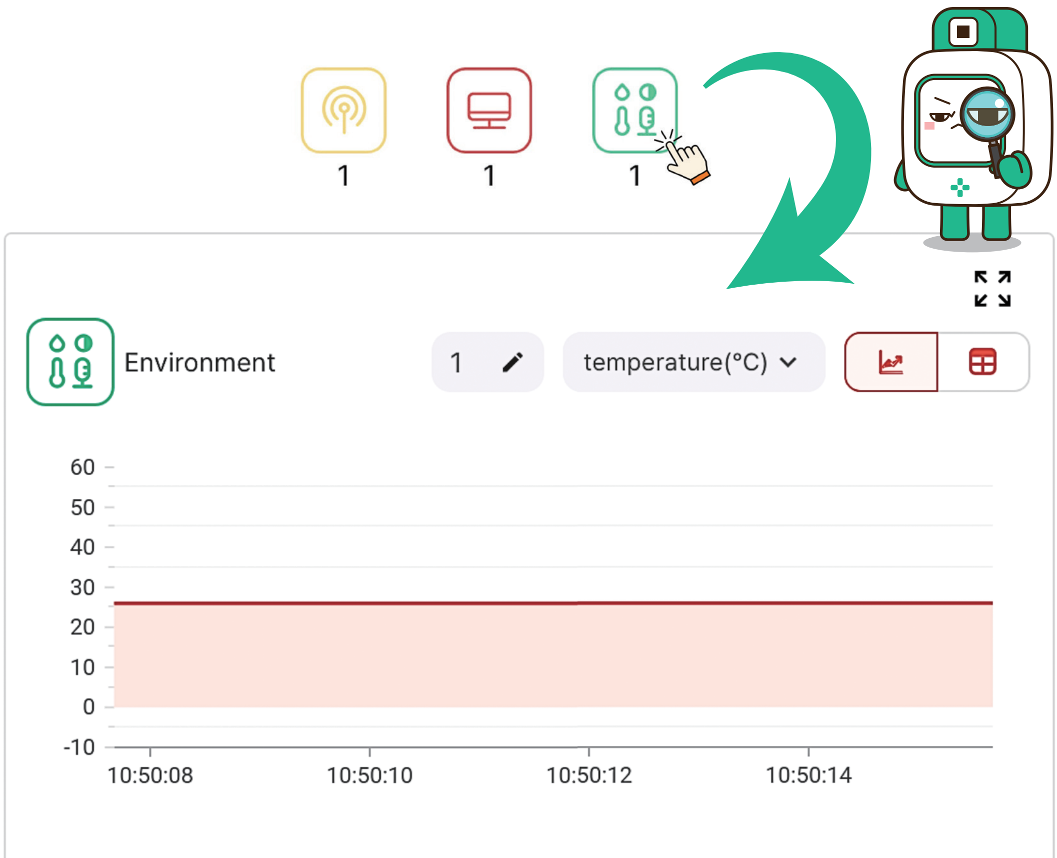
Display



Environment

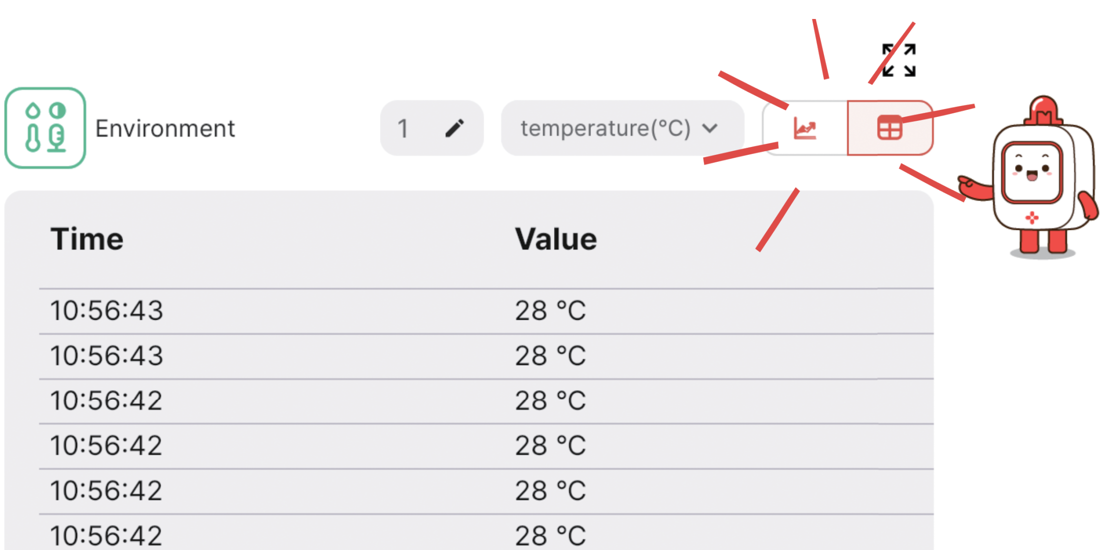
Network

To monitor the Environment module, click its icon in the module map.



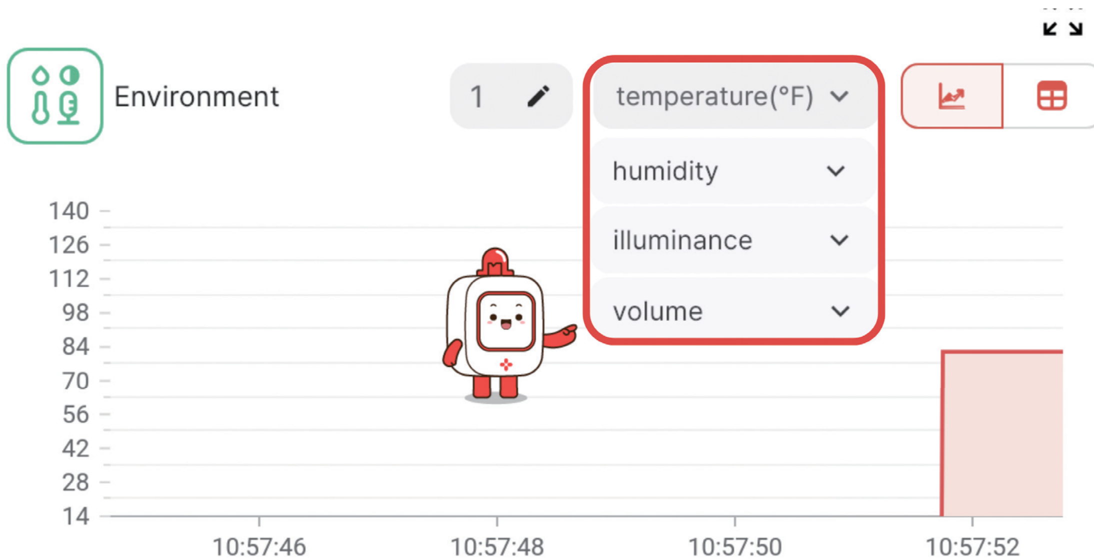
Monitoring Features:

- Switch between chart and graph views.



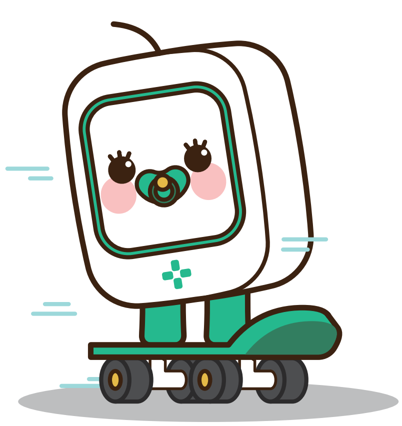
Time	Value
10:56:43	28 °C
10:56:43	28 °C
10:56:42	28 °C
10:56:42	28 °C
10:56:42	28 °C
10:56:42	28 °C

The Environment module can monitor four types of data, allowing temperature monitoring in both Celsius and Fahrenheit.

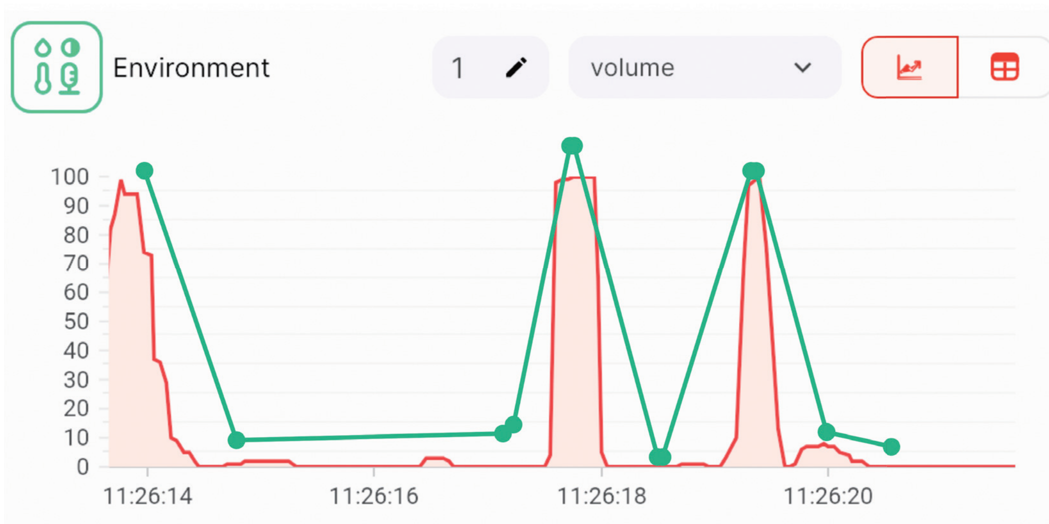


What is the temperature and humidity where you are?

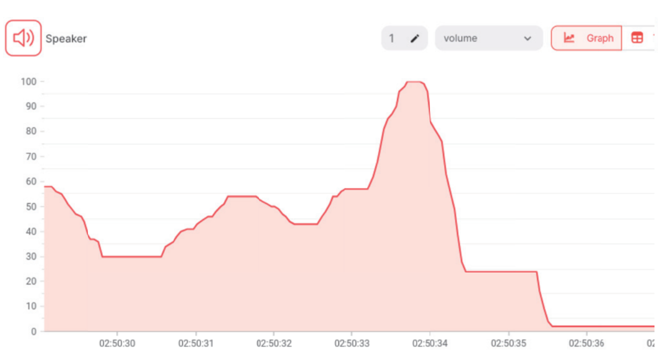
Temperature: **Humidity:**

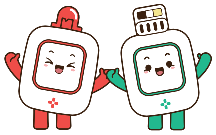


The data for illuminance and volume update in real-time, allowing you to observe changes directly on the graph and table. Try being as quiet as possible to see if we can get the volume level down to 0.



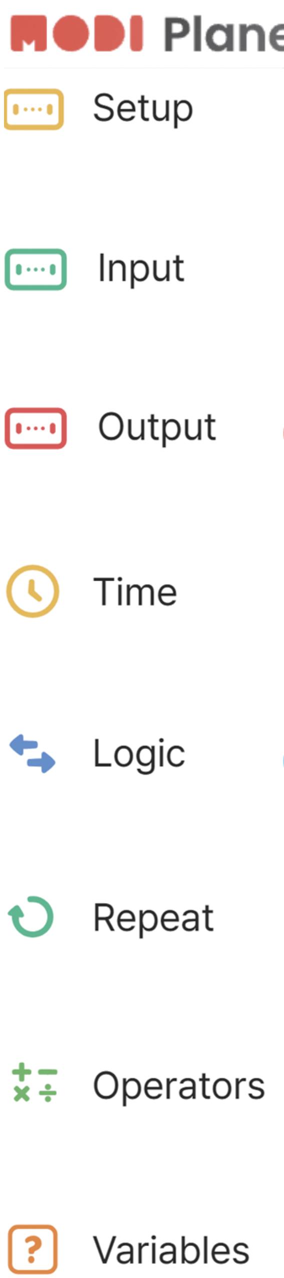
Both input and output modules can have monitoring capabilities, but not all output modules do.





TYPES OF BLOCKS IN THE BLOCK AREA

The Block Area is organized into seven tabs, each housing specific types of blocks:



1. **Network-Related Blocks:** These include the essential code start blocks that initiate your program.
2. **Input Module Blocks:** Tailored for interacting with various input modules.
3. **Output Module Blocks:** Designed to control and utilize output modules.
4. **Time Block:** Used to pause the code execution for a set duration.
5. **Conditional Statements:** Includes 'if' and 'if-else' blocks for making decisions.
6. **Loop Statements:** Contains blocks for repeating actions, like 'repeat'.
7. **Operators:** Focus on arithmetic calculations, comparisons, and generating random values.
8. **Variable Blocks:** Used for creating and modifying variables, crucial for storing and using data in your programs.

Blocks: Your First Step in Coding

Think of each block in the Code Editor as a stepping stone to becoming a coding pro. Just like these blocks, text coding involves giving commands to computers, but instead of blocks, you use written words.

NETWORK, INPUT, AND OUTPUT BLOCKS

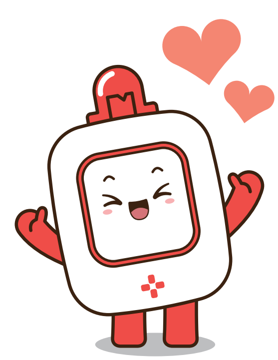
Starting things up with these blocks is just like using "go" commands in text coding. It's all about making actions happen!

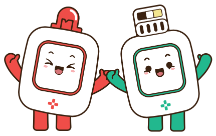
WAIT, CONDITIONAL, AND LOOP BLOCKS

Using these blocks to decide what happens next is the same as making choices and repeating actions in text coding. It's like deciding whether to jump or keep running in a game.

OPERATORS AND VARIABLE BLOCKS

Working with these blocks for numbers and information is exactly like solving puzzles with code words. It's how you keep score or remember important clues!

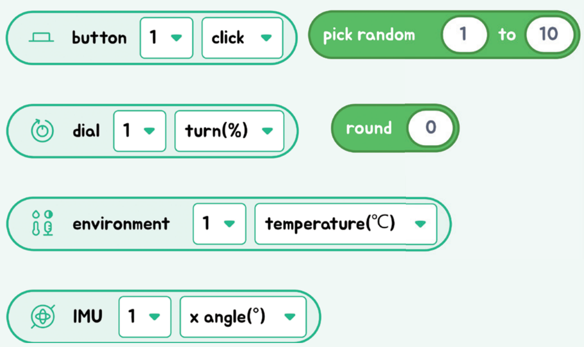




SHAPES OF BLOCKS

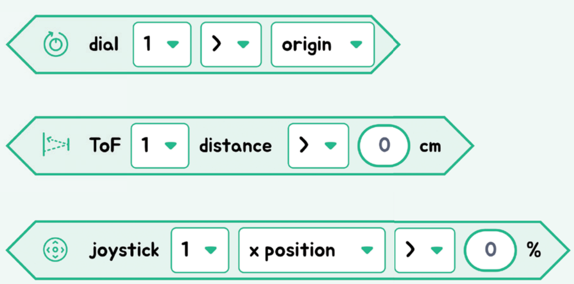
In Code Editor, aside from the network block that initiates your code, there are three distinct shapes of blocks.

1



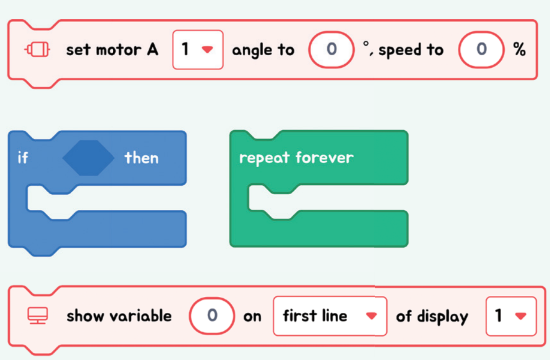
① Rounded Blocks:
These blocks hold values, like numbers. Think of them as small baskets carrying important information, usually numbers, for your code.

2



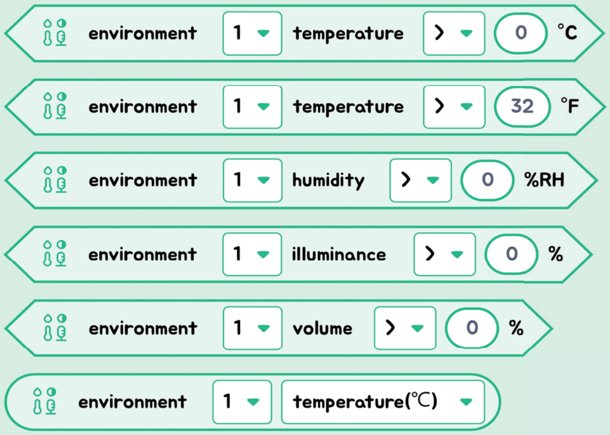
② Angled Blocks:
These blocks deal with conditions, asking "yes or no" questions. If the answer is "yes" (true), one thing happens; if "no" (false), something else might happen.

3



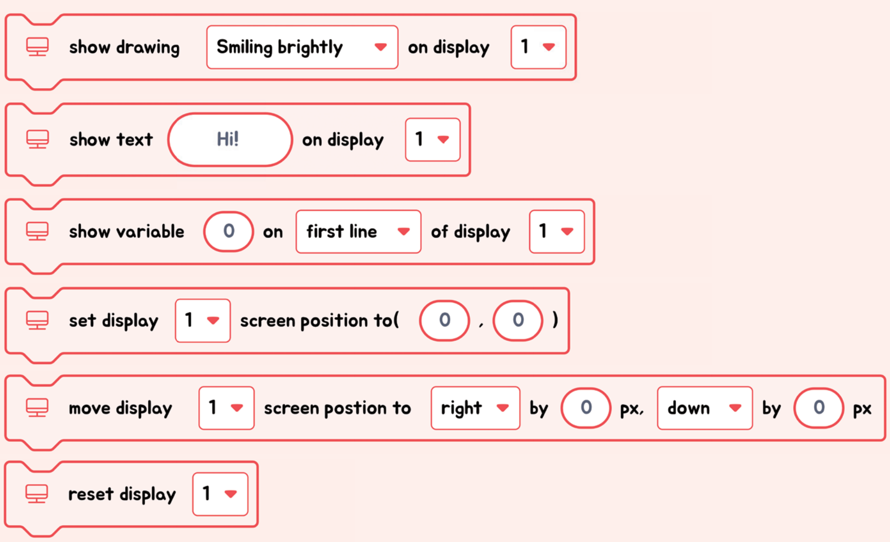
③ Rectangle-Shaped Blocks:
These square or rectangle blocks are all about action. They tell your MODI modules what to do, like moving, lighting up, or making sounds.

Blocks of Environment



The input module called Environment doesn't perform actions. Instead, it provides 5 condition blocks and 1 value block, each related to different types of environmental data it can sense. The condition blocks are used to check specific environmental conditions, and the value block is used to retrieve the exact measurements of those conditions.

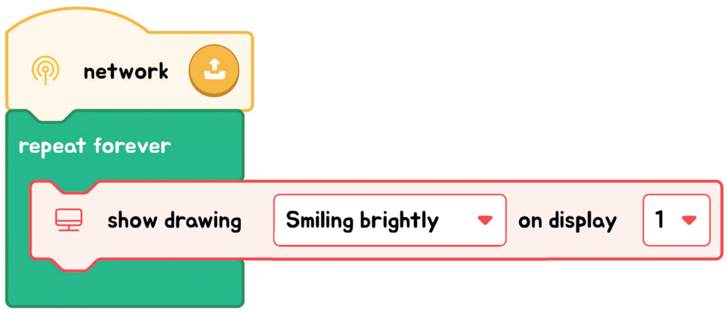
Blocks of Display



The output module, Display, is all about taking action. It has blocks specifically designed for showing pictures, text, or variable values on the screen. There are also blocks that allow you to move the displayed content around, and one important block that resets or initializes the Display for a fresh start.

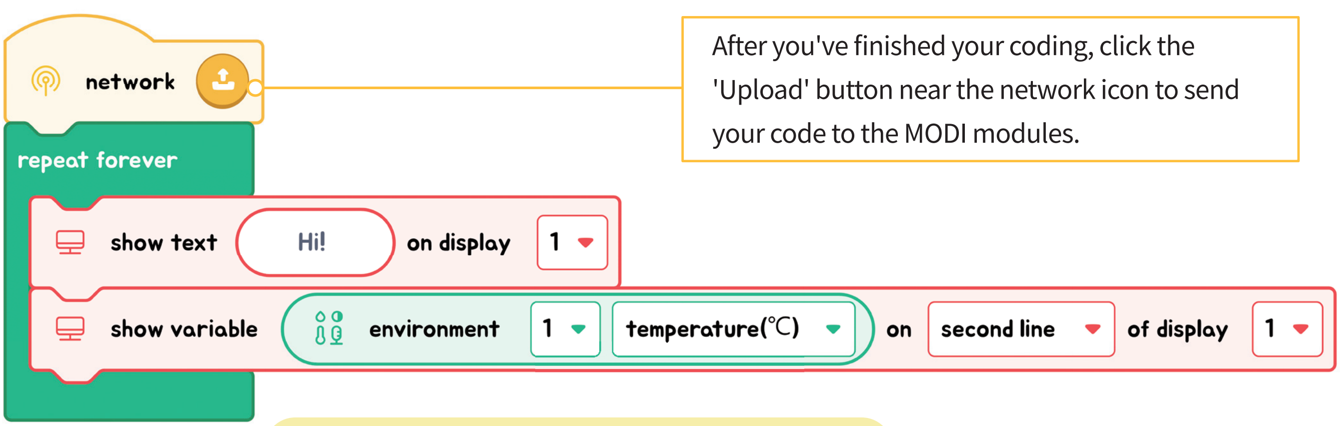
Showing a Picture on the Display

Let's get a picture to appear on the display screen. You can choose a different picture from the list and see how it looks!

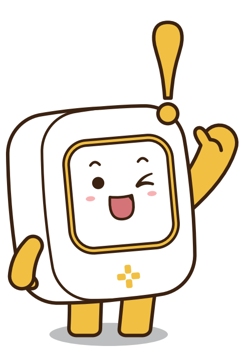


Displaying Values on the Display

Now, let's have the display screen show the word 'temperature' and then, right below it, show the actual temperature that the Environment module detects.



After you've finished your coding, click the 'Upload' button near the network icon to send your code to the MODI modules.



To prepare MODI for a new set of code, simply press the 'Initialize' button. You'll know MODI is ready when the status LED on the module turns blue.